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PAPER

Preliminary measurements of alpha particles produced by proton-boron nuclear fusion reaction on Z-pinch type discharges

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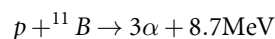
Abstract

The $^{11}\text{B}(p, \alpha)2\alpha$ fusion reaction occurring in hydrogen-boron fuel seems to be the most promising, therefore it is currently being investigated in various systems such as magnetic confinement fusion, inertial confinement fusion, high-power lasers and Z-pinch (Ribeyre *et al* 2022 *Sci. Rep.* **12** 4665), (Picciotto *et al* 2014 *Phys. Rev. X* **4** 031030–1). As this reaction does not produce neutrons and the energy released is carried only by alpha particles, it offers the possibility of direct conversion of kinetic energy into electricity (Lerner *et al* 2011 *J. Fusion Energy* **30** 367). The presented work demonstrates preliminary results of measurements of alpha particles produced by the aforementioned reaction in Z-pinch-type facilities. High-voltage, high-current discharges in Plasma Focus-type devices lead to the creation of a dense column of high-temperature magnetized plasma, which emits pulsed electron- and ion-beams, x-ray pulses etc. The results shown were obtained at the PF-360 and PF 1000 facilities, filled with H_2 – gas. The experimental campaign was conducted to test the possibility of producing intense beams of high-energy protons that can induce a considerable number of the $^{11}\text{B}(p, \alpha)2\alpha$ reaction (Scholz *et al* 2019 *J. Fusion Energy* **38** 522).

1. Introduction

The proton-boron fusion reaction, which was first conducted in 2005 [5], continues to be the subject of intensive research because of its widespread interest. Most research focuses on the application of reaction products, i.e. alpha particles with a wide energy spectrum, in medicine as a source for the production of radioisotopes, or directly in boron proton capture therapy [6]. In addition, its potential application lies in the field of controlled nuclear fusion reactions. As demonstrated in the literature [7], using an appropriate target it is possible to achieve a number of alpha particles well above 10^{10} $\alpha/\text{sr}/\text{shot}$ on targets irradiated at a laser intensity of about 3×10^{16} W cm^{-2} at PALS [8].

The main aim of this work was to attempt to produce a high yield of alpha particles by triggering the following proton–boron nuclear reaction,



and to measure the alphas using CR-39 TASTRAK detectors [9]. From this reaction channel, alpha particles with energy distribution ranging from 2.5 MeV to 5.5 MeV are generated. Their maximum energy is equal to about 4.3 MeV. It is well known (from theoretical calculations and experimental measurements) that the reaction has a main channel with a maximum cross section for protons with energies of 600–700 keV [10, 11].



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Polyallyl diglycol carbonate (PADC) detectors are widely used for dosimetry applications, such as air radon concentration dosimeters, neutron monitoring, and in space research to investigate the depth distribution of radiation in different organs of astronauts during long-duration missions [12, 13]. Samples of such detectors are also applied in different diagnostic tools and irradiated in various high-temperature plasma experiments (Plasma Focus (PF) [14], Tokamak [15], laser-produced plasma [2] and also in a medical accelerator) [6, 9]. Appropriately calibrated PADC detectors would help to make such measurements simpler and more precise, and thereby would cause these detectors to find applications in new research areas. In the described experiment, we tested one kind of PADC of CR-39 TASTRAK type detectors manufactured by TASL Analysis System Ltd (Bristol, United Kingdom).

2. Calibration of CR-39 TASTRAK detectors

To perform calibration measurements of the detectors used, some selected samples were placed in a vacuum chamber and irradiated with quasi mono-energetic hydrogen and helium ions of energies from 0.1 MeV up to 2.0 MeV, provided by an electrostatic Van der Graaf accelerator. During irradiation the energy values of proton and alpha were changed by 200–300 keV for each calibration point. Additionally, 4He^+ ions of energies in the range 2–5 MeV were obtained by using an ^{241}Am radioisotope source and degrading the primary α particle energy by a 4.5 mg cm^{-2} thick copper foil. The flux of ions reaching the track detectors was below $(2\text{--}4) \times 10^4\text{ particles cm}^{-2}$. It was estimated on the basis of measurements with a semiconductor detector, and adjusted to the aforementioned values during calibration. The samples after irradiation were chemically etched in 1 h-steps in a 6.25 N NaOH solution at a temperature of $70^\circ \pm 1^\circ\text{C}$ and then washed and dried. The readout of calibration samples was made by means of the optical microscope coupled with automated XY-stage and CCD camera, which were connected to a PC with suitable software. To analyze the etching tracks, we used python and ImageJ [16]. Obtained data show that the value of the track diameter rapidly increases in the region of relatively low energies. After reaching the maximum, the track diameter D , monotonically decreases with the energy (see e.g. figure 3). In this region of D versus ion energy E , the $D(E, t)$ graph approximately corresponds to the stopping power curve. We also plotted the distributions of the mean gray level values for these ions (figure 2).

The selected results of our calibration measurements are presented in figures 1 and 2. The figures present energy spectra and two-dimensional distributions of track mean gray level value versus track diameter as determined for 3 h etching time for proton and alpha ions. Using the data from both the track diameter diagrams and the mean gray level dependencies we are able to correctly estimate the particle energy (figure 1). Additionally, we are able to separate ions with lower energy from those with higher energies, producing tracks with the same diameter but with various mean gray levels (see figure 2, e.g. alpha particle tracks with 10 μm diameter).

Based on extensive experience, it is recommended that a newly used part of such detectors be calibrated due to the lack of data from the manufacturer regarding changes in detector production. Therefore, some rectangular samples of the CR-39 type detectors that were used in the present experiment were irradiated with 5 MeV α particles from an ^{241}Am source. Lower ranges of energies were obtained using an Al foil of different thicknesses. Figure 3 presents a calibration curve for alpha ions in a given energy range.

3. Experimental procedures

Research on the emission of alpha particles generated in $p+^{11}\text{B}$ reaction presented here was performed within the PF-360 facility located in NCBJ, Świerk [14] and within the DPF-1000U facility, Warsaw [17].

The time-integrated measurements of alpha particles were carried out for a chosen series of PF discharges under the experimental conditions, as follows: $p_0 = 11\text{--}12\text{ hPa}$ H_2 , $U_0 = 30\text{ kV}$ in the case of PF-360, or $p_0 = 1.81\text{ Tor}$ H_2 , $U_0 = 16\text{ kV}$ in the case of DPF-1000U. In our laboratory, we conducted extensive research involving measurements of primary protons on the PF-type facility of various sizes. Our experience shows that the angular distribution of primary protons, which was determined for the PF, is strongly anisotropic. This means that the recorded proton flux depends strongly on the observation angle measured in relation to the electrode axis and the maximum emission is located at small angles. This angle has a typical value up to 10 degrees with respect to the electrode symmetry axis [18]. For this we also used the CR-39 detector, in which samples were placed on a special semi-circular support located inside the PF-1000 facility discharge chamber, at a distance of 300 mm from the inner electrode outlet center (figure 4). The SRIM program [19] was used to obtain the most appropriate position of the track detector relative to the particle generation source at known experimental conditions (i.e. the

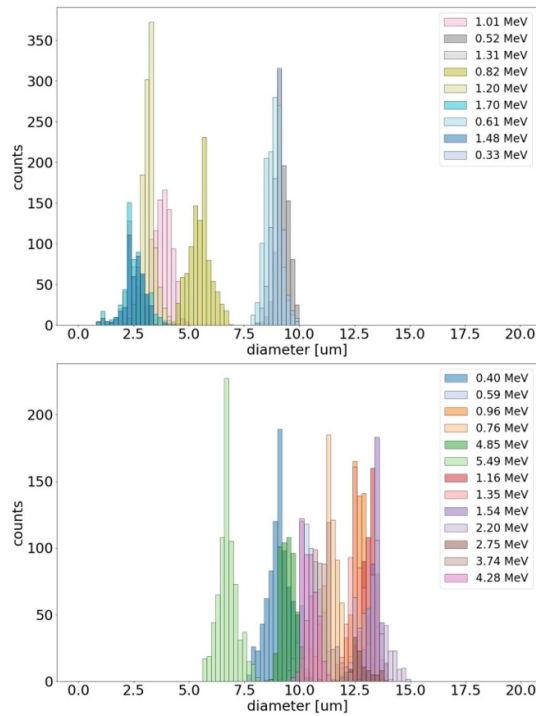


Figure 1. Energy spectra of CR-39 TASTRACK detector as determined for 3 h etching time for proton (left) and alpha ions (right).

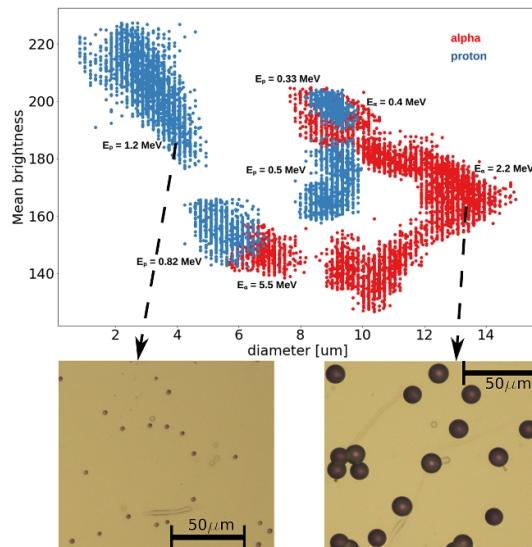


Figure 2. Two-dimensional distributions of track mean gray level value versus track diameter as determined for 3 h etching time.

pressure, type of gas and the energy of the particles). The program calculates the stopping power and range of ions during their motion through a gas-filled region. After irradiation, the detector samples were etched under the same conditions as the calibrated samples. Some pictures of tracks produced by primary protons and recorded on the CR-39 detector samples are presented in figure 5.

Such a large anisotropy cannot be explained only by the influence of local (azimuthally- and axially-oriented) magnetic and electric fields on primary proton trajectories, but may also be a result of stronger deflection of ions due to higher values of the total current, flowing through the pinch. It should also be noted that the angular distributions shown in many papers [18, 20] demonstrated axial symmetry to the electrode axis.

Figure 6 presents the energy spectrum of the primary protons, which were determined within the DPF-1000U experiments. A diagram presenting the number of recorded primary proton tracks versus

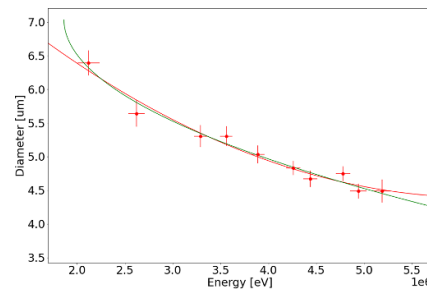


Figure 3. Diameter versus energy of alpha particle tracks, after 3 h etching time.



Figure 4. Special holder equipped with ion pinhole cameras, used for determination of angular distribution of primary protons.

the proton energy is obtained on the basis of the calibration curves. The red line represents the $p+^{11}\text{B}$ reaction curve. The particles were recorded along the electrode symmetry axis at about 0 degrees, where the maximum number of protons was observed. In order to estimate the energies of the considered primary protons leaving the plasma column one should take into account the energy loss during their motion through a gas-filled region. On this basis, in our measurements the distance from the plasma pinch to the detectors was fixed at 30 cm.

The registered primary protons, which were emitted from hydrogen discharges of the DPF-1000U device, had energies within the range of 600 keV to about a few MeV. A maximum spectrum is obtained in the energy range from 600 to 800 keV. One can see that the DPF-1000U device emits ions with energies even greater than 1 MeV despite a much lower energy discharge (charging voltage). It is clearly seen that the observed spectra have two distinguished components: one is faster disappearing with energy range 0.65–0.8 MeV and the other is slower in the area above 0.8 MeV. These spectra are similar to those in the experiment at other PF facilities.

4. Results and discussion

The alpha particles were measured using a PADC detector placed in close vicinity of the boron target. The samples of the track detectors, as well as the pure boron target, were placed on a special holder located inside the vacuum chamber, at a chosen distance from the inner electrode outlet center on the axis (figure 7).

The tracks corresponding to alpha particles were recognized on the basis of the performed detector calibration. In order to reduce the impact of noise on detector readout, we have applied a set of filters, based on track properties. The proper alpha track had to have a brightness and diameter within a certain range, as well as being characterized by a smooth, convex border. We have also rejected tracks with a too high aspect ratio (i.e. ratio of major to minor axis of ellipse), since a too high eccentricity does not allow proper calculation of track diameter. Applying the aforementioned filters (diameter and brightness within a certain range, circularity and aspect ratio above 0.75 and solidity above 0.8), we were able to increase the signal-to-noise ratio of our detectors.

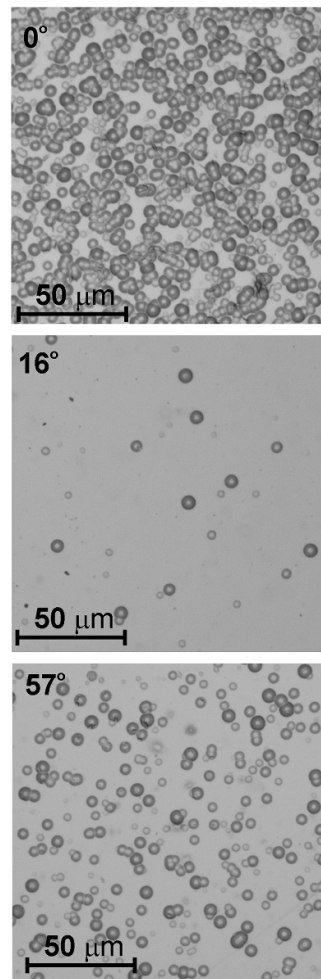


Figure 5. Example pictures of proton craters, which were recorded upon detector samples located at various angles to the DPF-1000U device electrode axis.

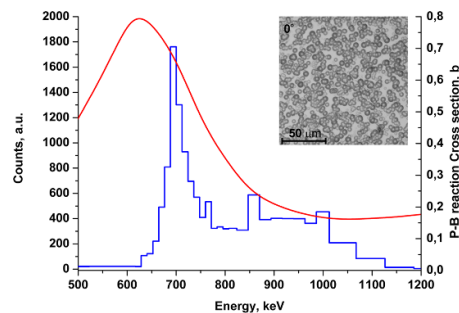


Figure 6. Energy spectrum of primary protons (emitted along the axis), which was determined within the PF experiments (left axis). The red line represents the $p+^{11}\text{B}$ reaction cross section (right axis).

The results, presented in figure 8, show that the CR-39 detector placed inside the DPF-1000U vacuum chamber registered alpha particles. The number of alpha tracks is clearly greater than the background level, and the tracks are distributed asymmetrically, which could be the result of the relative position of the boron target and the CR-39 detector.

Finally we estimated that at the current experiment state the number of alpha particles reaching the track detector surface is about 120 per square centimeter. Since the total number of alphas was very small, we decided that further investigation and improvements were necessary. On the basis of the conclusions mentioned above, we performed experimental measurements of alpha particles during irradiation of the new-type boron target.

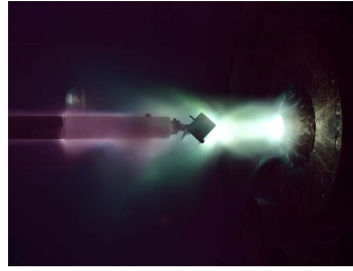


Figure 7. Side-view of the coaxial electrodes and a special holder, which was used for measurements of alphas particles.

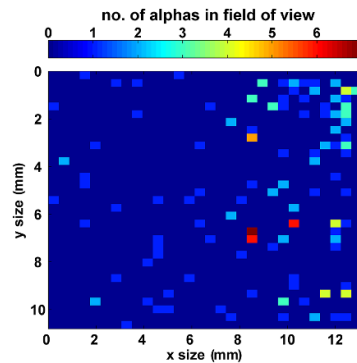


Figure 8. Distribution of tracks over the detector surface revealed in the microscope field of view.

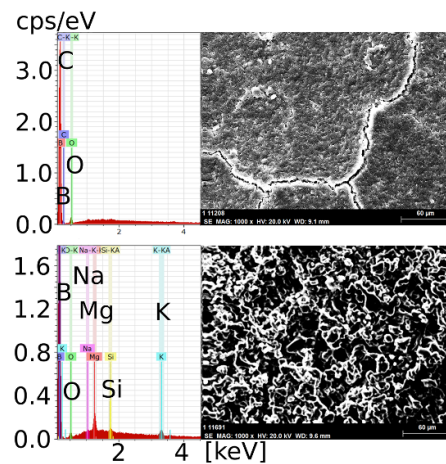


Figure 9. EDS analysis of surface of boron targets used in PF experiments. Old-type (top) and new-type of target (bottom).

Figure 9 shows energy-dispersive x-ray spectroscopy (EDS) analysis of the surface of boron targets used in PF experiments: old-type (top) and new-type target (bottom). The chemical composition analysis revealed significant differences in the quality of the initial material. In the case of the old-type target, the presence of oxygen was detected in its composition. Partial oxidation of the powder surfaces and insufficient compaction density led to the formation of microcracks and the development of an oxidized layer during the experiment. These phenomena had a negative impact on the system stability and reduced the efficiency of alpha particle generation. The introduction of the new-type target, characterized by higher surface purity and the absence of oxygen-related contaminants, improved the experimental stability and increased the yield of alpha particle emission.

New target types have been used at the PF-360 facility. The track detector was placed on a holder which was used to place boron material into the vacuum chamber. Similarly to the previous experimental session, the track detector was placed on a surface perpendicular to the target plane, about 2 cm from the target center. The distance between target and anode end was equal to 20 cm. We estimated

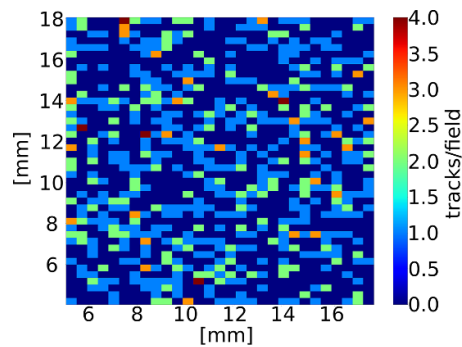


Figure 10. Distribution of tracks over the detector surface revealed in the microscope field of view.

that at the present experiment state the number of alpha particles reaching the track detector surface is about 440 particles per square centimeter, which is considerably greater than the background (50 tracks per sqcm) (figure 10).

5. Conclusions

The use of the determined calibration curves and mean gray level values makes possible the conversion of track histograms into energy spectra of the measured ions. It was shown that the CR-39 detectors are very useful diagnostic tools for measurements of primary protons, emitted from high-temperature hydrogen plasmas produced by PF-type discharges, as well as alpha particles. The energetic primary protons, which were recorded during hydrogen discharges at the PF facilities, had energies ranging from 600 keV to a few MeV. We note that the $p+^{11}\text{B}$ reaction requires an average ion energy above 100 keV to 900 keV (maximum 650 keV, figure 6, red line) which coincides with the received spectra. The number of alphas was not very large at the present state of the Plasma Focus experiment and further investigations and improvements are necessary.

It should be noted that the use of a pure metallic boron target allowed us to record a significantly greater number of alpha particles.

Further investigation of irradiation geometry, as well as numerical analysis of plasma-target interaction are needed for optimization and increase in fast alpha particle generation.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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